

Project Report : A study of the limits of Fiberwise Dimensionality Reduction with Vector Bundles.

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Abstract

Datasets are becoming increasingly bigger and more complex, making direct analysis, visualization, and learning both computationally expensive and conceptually difficult. Hence the importance of dimension reduction algorithm. In this project we study a recently proposed dimensionality reduction algorithm that models complex datasets as vector bundles. We propose to focus on the limitations and improvements given by the authors at the end of their paper. Mainly, we aim to show that data containing outliers can significantly affect the quality of the algorithm as only a subset of the data satisfies the assumptions in this case.

1 Introduction.

The main idea behind the paper is that when considering a dataset as a finite sample of a Riemannian manifold, one should not focus only on preserving distances, but instead aim to produce a low-dimensional embedding that captures both the local geometry and global topology.

Most common dimensionality-reduction methods are called metric-based (e.g Isomap, Local Tangent Space Alignment (LTSA), and Hessian Eigenmaps (HLE)). They try to preserve distances between pairs of points. However these methods fail when manifolds have non-trivial global topology (e.g. a circularity, like a cylinder projected in $\mathbb{R}^2$). The contribution of the authors is the use of vector bundles, i.e a topological base carrying global structure and attached fibers carrying local geometry, in order to model complex data. This method separates global topological features from local geometric variation. In practice, this allows the

algorithm to compress the dimension without forcing a single global metric-preserving embedding (which may not exist in low dimension).

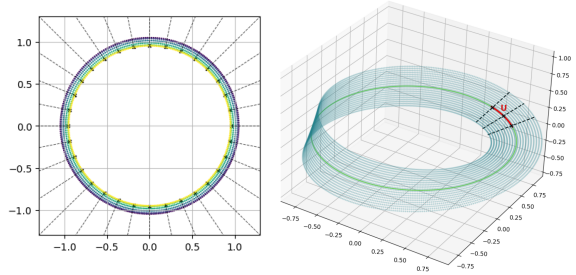


Figure 1: Visualization of the base and fibers of the cylinder ($S^1 \times \mathbb{R}$) and the moebius strip (a line bundle over S^1). In the left image, the base corresponds to the unit sphere in dimension 1 S^1 while the fibers correspond to the vertical lines $\mathbb{R}$ glued to each point of the sphere (here the fibers stop at 0.15 of the radius of the cylinder). In the right image, locally around every point of S^1 it looks like $U \times \mathbb{R}$ where U is an open arc, the total bundle is different from $S^1 \times \mathbb{R}$ which is the cylinder.

2 Underlying Cohomology and Intrinsic Dimensionality.

A recurring observation in data analysis is that high-dimensional datasets rarely occupy the full ambient space $\mathbb{R}^D$ in which they are represented. Instead, they typically concentrate on a much smaller subset, a manifold or a stratified space, whose intrinsic dimension is often far lower than D . Identifying this intrinsic structure is essential for meaningful dimensionality reduction.

Cohomology offers an algebraically viewpoint that is often flexible for constructing coordinates. In particular, the first cohomology group H^1 detects the presence of cycles, nontrivial one-dimens

ional loops along which the data evolve. Such cycles arise in many practical settings, for instance in periodic physical systems, biochemical reaction coordinates, or datasets exhibiting rotational or cyclic behavior.

In the context of data, the key insight is that a nontrivial cohomology class in H^1 can be used to produce a global coordinate on the data manifold. Persistent cohomology strengthens this idea by providing a multiscale description, allowing us to detect which topological features persist across noise and across different resolutions of the data. When a robust one-dimensional cohomology class is present, it can be converted into a *circular coordinate*, that is, a map

$$\theta : X \longrightarrow S^1 \quad (1)$$

which parametrizes the cycle underlying the dataset.

This perspective shows that many datasets admit an effective one-dimensional representation when a persistent class in H^1 is present; more generally, cohomology can reveal low-dimensional intrinsic structure whenever higher-degree groups such as H^2 or H^3 contain meaningful features. Thus, even though the data may be embedded in a high-dimensional ambient space $\mathbb{R}^D$, they often lie on a much smaller subset whose global topology is encoded by their cohomology. Cohomological coordinates make this intrinsic structure explicit. Detecting and exploiting this structure is precisely what cohomological methods make possible.

To illustrate these ideas, we introduce a simple example in dimension three. The choice of $\mathbb{R}^3$ is purely practical: it allows for visual representation while keeping the underlying topology nontrivial. The dataset we construct does not fill the entire cube $[0, 1]^3$ but instead lies on a closed one-dimensional curve inside it.

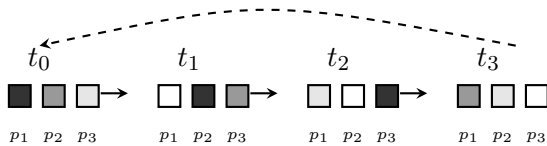


Figure 2: Schematic representation of cyclic data using pixels in grayscale in $\mathbb{R}^3$

Cohomology detects this curve as a one-dimensional cycle, enabling a meaningful reduction from

dimension three to dimension one. The configurations in figure 2 illustrate successive states of the three pixels at different instants. As the intensity pattern rotates among the pixels, the system follows a closed loop in the three-dimensional intensity space, which is precisely the one-dimensional cycle detected by cohomology.

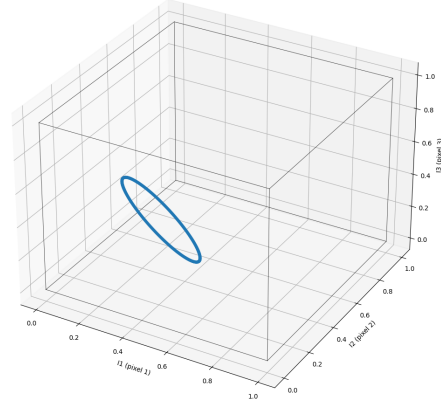


Figure 3: Illustration of the cycle the pixels follow in $\mathbb{R}^3$

The example was purposely constructed in $\mathbb{R}^3$ for visualization purposes, yet nothing in principle restricts the approach to low dimensions. The same procedure, detecting a persistent class in H^1 and turning it into a circular coordinate, can be carried out in any $\mathbb{R}^n$, as described in the algorithm presented in the paper (Topological Data Analysis). Cohomology effectively extracts the base space of the data.

Nevertheless, this purely cohomological viewpoint is only suitable when the data are clean and exhibit a nearly perfect cyclic structure. It ignores any additional local geometry the dataset may possess, since a single circular coordinate cannot encode variations transverse to the cycle. In more realistic settings where the data may have curvature, local degrees of freedom, or fiber structure, one needs a richer model. This is precisely the motivation behind the fiberwise dimensionality reduction framework: to retain the global topology while faithfully capturing local geometric information. Although the FibeRed algorithm takes this topological base as given, understanding how it arises is essential for grasping the full picture.

3 Vector Bundles and Fiberwise Dimensionality Reduction.

Geometric intuition : A vector bundle models data as having two distinct components: a base space B encoding the global topological structure, and fibers attached to each point of B encoding local geometric variations.

Consider a cylinder, embedded in $\mathbb{R}^3$. Its base space $B = S^1$ is a circle capturing the essential circularity of the data. And at each point $b \in S^1$, we attach a vertical line (the fiber) $\pi^{-1}(b) \simeq \mathbb{R}$. The projection map $\pi : S^1 \times \mathbb{R} \rightarrow S^1$ sends $(b, t) \mapsto b$, "collapsing" each fiber to its base point. The cylinder is thus the total space $X = S^1 \times \mathbb{R}$, the union of all fibers parametrized continuously by the base.

For dimensionality reduction, if we can represent the base B explicitly (e.g., embedding S^1 as a circle in $\mathbb{R}^2$) and compress each fiber from ambient dimension D to intrinsic dimension d , then we obtain an embedding in $\mathbb{R}^{\dim(B)+d}$ rather than the potentially much larger space required for global isometry.

For a more formal definition, a vector bundle E of rank d consists of three objects :

1. A total space X
2. A base space B
3. A continuous surjective map $\pi : X \rightarrow B$ called the projection

For each point $b \in B$, the preimage

$$\pi^{-1}(b) = \{x \in X : \pi(x) = b\} \quad (2)$$

is called the fiber above b and is required to be a real vector space of dimension d .

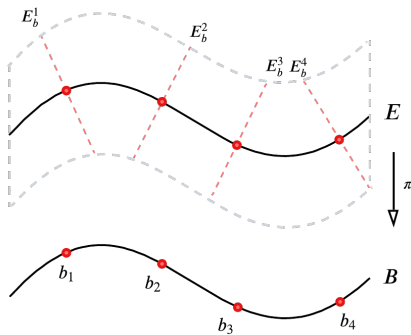


Figure 4: Schematic of a vector bundle E and a base B . The points b_i , $i \in \mathbb{N}$ correspond to the origin of the fibers E_b^i , $i \in \mathbb{N}$, and are mapped down by the projection π .

It is important to note that the bundle is locally trivial, meaning for every point $b \in B$, there exists an open neighborhood $U \subset B$ containing b and a homeomorphism $\varphi_U : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^d$ called a local trivialization. This trivialization must satisfy two compatibility conditions.

1. $\text{proj}_1 \circ \varphi_U = \pi$, meaning the trivialization respects the projection structure.
2. For each $b' \in U$, the restriction :

$$\varphi_U|_{\pi^{-1}(b')} : \pi^{-1}(b') \rightarrow \{b'\} \times \mathbb{R}^d \simeq \mathbb{R}^d \quad (3)$$

must be a linear isomorphism of vector spaces.

Intuitively, this means that locally over U , the bundle looks like the simple product $U \times \mathbb{R}^d$, but globally the fibers may be "twisted" in a way that cannot be captured by a single global coordinate system.

Vector bundles are obtained by gluing together local products $U \times \mathbb{R}^d$ using transition functions, which tell us how to match fibres above overlapping regions, and the cocycle condition, which expresses the fact that these matchings must agree when three patches meet.

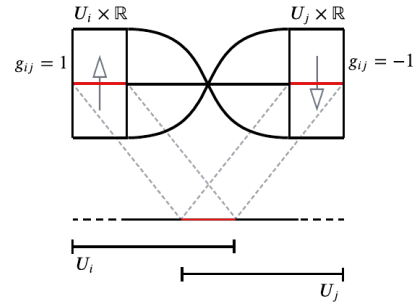


Figure 5: Schematic of the construction of a moebius strip on open sets U_i and U_j of the circle S^1 . It is obtained by using the non-trivial gluing of $g_{ij} = 1$ on one overlap and $g_{ij} = -1$ on the other.

When two open sets U_i and U_j overlap, the bundle $\pi^{-1}(U_i \cap U_j)$ can be identified with $(U_i \cap U_j) \times \mathbb{R}^d$ using the transition function

$$g_{ij} : U_i \cap U_j \rightarrow \text{GL}(d, \mathbb{R}) \quad (4)$$

this function measures how these coordinate systems relate to each other. Explicitly, for each $b \in U_i \cap U_j$, we have a linear isomorphism $g_{ij}(b) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ defined by $g_{ij}(b) = \varphi_i \circ \varphi_j^{-1}|_{\{b\} \times \mathbb{R}^d}$. These transition functions need to satisfy the cocycle conditions, ensuring consistency :

1. A trivialization agrees with itself :

$$g_{ii}(b) = \text{Id}, \quad \forall b \in U_i \quad (5)$$

2. Compatibility on triple overlaps :

$$g_{ij}(b) \cdot g_{jk}(b) = g_{ik}(b), \quad \forall b \in U_i \cap U_j \cap U_k \quad (6)$$

3. Transitions are invertible :

$$g_{ij}(b) = g_{ji}(b)^{-1} \quad (7)$$

When the bundle carries a Euclidean structure, an inner product on each fiber varying smoothly with the base point, we can choose trivializations such that all transition functions take values in the orthogonal group $O(d)$. This means that $g_{ij}(b)$ is not just a linear isomorphism but an isometry, preserving the geometric structure of fibers. The collection of transition functions $\{g_{ij}\}$ is called a cocycle and completely determines the bundle up to isomorphism. Different cocycles correspond to different "twistings" of the fiber structure over the base.

Suppose we are given an embedding $\iota : B \rightarrow \mathbb{R}^D$ of the base space that captures its essential topology (for instance, embedding a circle as the unit circle in $\mathbb{R}^2$). The vector bundle embedding problem asks :

Can we extend this to an embedding $\tilde{\iota} : X \rightarrow \mathbb{R}^D$ of the entire bundle in a way that respects the bundle structure ?

More precisely, we seek a map $\tilde{\iota} : X \rightarrow \mathbb{R}^D$ satisfying three properties.

1. $\tilde{\iota}$ extends ι : meaning if $s_0 : B \rightarrow X$ denotes the zero section (sending $b \mapsto$ the origin of the fiber $\pi^{-1}(b)$), then $\tilde{\iota} \circ s_0 = \iota$.
2. $\tilde{\iota}$ is fiberwise isometric meaning the restriction to each fiber $\pi^{-1}(b)$, $b \in B$ is a linear isometry.
3. The embedded fibers are orthogonal to the embedded base : $\tilde{\iota}(\pi^{-1}(b)) \perp \iota(T_b B)$ for all $b \in B$.

This problem can be reduced to pure linear algebra on cocycles. Let $N \rightarrow B$ denote the normal bundle of $\iota(B) \subset \mathbb{R}^D$, the bundle whose fiber at b is the orthogonal complement of $\iota(T_b B)$ in $\mathbb{R}^D$. The embedding $\tilde{\iota}$ exists if and only if there exists a bundle morphism $X \rightarrow N$ that is fiberwise isometric. In terms of local trivializations, this amounts to finding orthonormal frames

$$\Phi_i : U_i \rightarrow V(r, D - e) \quad (8)$$

where e is the dimension of B , r the rank of X , and $V(r, D - e)$ is the Stiefel manifold of r -frames in $\mathbb{R}^{D-e}$ such that

$$\Phi_i(b) \Omega_{ij}(b) = \Theta_{ij}(b) \Phi_j(b), \quad \forall b \in U_i \cap U_j \quad (9)$$

where $\{\Omega_{ij}\}$ is the cocycle of the data bundle X and $\{\Theta_{ij}\}$ is the cocycle of the normal bundle N .

This equation expresses a global alignment condition. The local fiber coordinates must be rotated by Φ_i in such a way that the data bundle's twist Ω_{ij} matches the twist Θ_{ij} of the normal bundle. Finding such $\{\Phi_i\}$ is the core mathematical problem solved by the FibeRed algorithm.

A major problem of this field of study is going from continuous to discrete. In practice, we do not have access to smooth manifolds and continuous functions. Instead, we have a finite point cloud $\mathcal{X} \subset \mathbb{R}^D$ sampled from an unknown manifold X and a map $f : \mathcal{X} \rightarrow B$ (the base projection, typically obtained from topological methods like circular coordinates). The challenge is that cocycles are defined as continuous functions $g_{ij} : U_i \cap U_j \rightarrow O(d)$, which cannot be represented exactly with finite data.

The solution introduced by Scoccola and Perea is to use discrete approximate cocycles. Instead of continuous maps, we represent each transition function by a single constant matrix $\Omega_{ij} \in O(d)$. This matrix captures the "average" rotation between the local fiber coordinates on patches U_i and U_j . The cocycle conditions then become :

1. $\Omega_{ij} = \Omega_{ji}^T$, since orthogonal matrices have transpose equal to inverse.
2. $\Omega_{ij}\Omega_{jk} \approx \Omega_{ik}$ for triple overlaps.

Given finite samples $\mathcal{X}_i \subset U_i$ and $\mathcal{X}_j \subset U_j$ with local fiber coordinates $\{f_i(x)\}_{x \in \mathcal{X}_i}$ and $\{f_j(x)\}_{x \in \mathcal{X}_j}$, the matrix Ω_{ij} is computed by solving the orthogonal Procrustes problem:

$$\Omega_{ij} = \arg \min_{Q \in O(d)} \sum_{x \in \mathcal{X}_i \cap \mathcal{X}_j} \|Q f_j(x) - f_i(x)\|^2 \quad (10)$$

This least-squares problem has a closed-form solution via singular value decomposition, making it computationally efficient. The discrete cocycles $\{\Omega_{ij}\}$ retain the essential algebraic structure of the continuous bundle while being computable from finite data, enabling practical implementation of the vector bundle embedding framework.

4 The FibeRed Algorithm.

The FibeRed algorithm takes as input a finite point cloud $\mathcal{X} \subset \mathbb{R}^D$ together with a base map $\pi : \mathcal{X} \rightarrow B$ and outputs an embedding that integrates local fiber structure with global base topology. The pipeline consists of four stages: covering the base space, extracting local fiber coordinates, estimating cocycles, and aligning fibers globally.

Algorithm 1 FibeRed($X, \pi, k, e, d, c, \text{n_iter}$)

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1:  $\mathcal{U}, \rho \leftarrow \text{Cover\&PartitionUnity}(k, B)$ 
2:  $\mathcal{N} \leftarrow \text{Nerve}(B, \mathcal{U})$ 
3: for  $1 \leq i \leq k$  do
4:    $l_i \leftarrow \text{LocalLinearRep}(X, \mathcal{U}, d, i)$ 
5:    $\Psi_i, \alpha_i \leftarrow \text{Est\&NormBun}(B, \mathcal{U}, e, i)$ 
6:    $\bar{f}_i \leftarrow \text{EstNormFiberCoord}(B, \Psi_i, l_i)$ 
7: for  $(i, j) \in \mathcal{N}$  do
8:    $\Omega_{ij}, \Theta_{ij} \leftarrow \text{ECocycles}(\bar{f}_i, \bar{f}_j, \alpha_i, \alpha_j)$ 
9:  $\tau \leftarrow \text{EstReach}(B, \mathcal{U}, \Psi)$ 
10:  $\Phi \leftarrow \text{AlignFibers}(\mathcal{N}, \Omega, \Theta, \text{n\_iter})$ 
11: return Assemble( $\rho, \tau, c, \alpha, \Phi, \bar{f}, \pi$ )

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Covering of the base space.

The algorithm first covers the base $B = \pi(\mathcal{X})$ using a greedy k -center algorithm to obtain centers $\{b_1, \dots, b_k\}$ and open sets $\{U_1, \dots, U_k\}$. The nerve $\mathcal{N}$ is the weighted graph with vertices corresponding to patches and edges (ij) weighted by $s_{ij} = |U_i \cap U_j|$ whenever patches overlap.

Extraction of the local fibers.

For each patch U_i , the algorithm extracts the local data $\mathcal{X}_i = \pi^{-1}(U_i)$ and builds local linear approximations.

Then the algorithm does a Principal Component Analysis on U_i with target dimension e yielding an orthonormal matrix Ψ_i approximating the tangent space to B at b_i , with normal space spanned by α_i .

Then a multidimensional scaling to $\mathcal{X}_i$ with target dimension d gives a local embedding $\ell_i : \mathcal{X}_i \rightarrow \mathbb{R}^d$. To extract fiber coordinates, the algorithm finds a linear map $m_i : \mathbb{R}^d \rightarrow \mathbb{R}^e$ that best predicts the base projection, then takes η_i as an orthonormal basis for $\ker(m_i)$. The fiber coordinate

map is $f_i = \eta_i^T \circ \ell_i$, normalized to lie in the unit ball.

Cocycles Estimation.

For each edge $(ij) \in \mathcal{N}$, the algorithm estimates the discrete cocycle $\Omega_{ij} \in O(r)$ of the data bundle by solving the orthogonal Procrustes problem, which is finding the optimal rotation aligning fiber coordinates on the overlap $\mathcal{X}_i \cap \mathcal{X}_j$. Similarly, the algorithm computes $\Theta_{ij} \in O(D - e)$ for the normal bundle cocycle by aligning the normal frames α_i and α_j . Both problems admit closed-form solutions via singular value decomposition.

The core task is to find orthonormal frames $\{\Phi_i \in V(r, D - e)\}$ minimizing the weighted alignment error :

$$\sum_{(ij) \in \mathcal{N}} s_{ij} \|\Phi_i \Omega_{ij} - \Theta_{ij} \Phi_j\|_F \quad (11)$$

When $D = r + e$, the Stiefel manifold $V(r, r) = O(r)$ is disconnected, so a $\mathbb{Z}_2$ -synchronization is performed using spectral methods to resolve sign ambiguities, then a restriction optimization to the connected component $SO(r)$ is done. The non-convex optimization is solved via stochastic gradient descent on the Stiefel manifold: at each iteration, an edge is sampled, the locally optimal alignment is computed, and the frames are updated using a decreasing learning rate.

Fiber Alignment & Assembly.

Finally the algorithm estimates the reach τ of the embedded base from pairwise distances between cover centers and their projections onto estimated tangent spaces. The final embedding is assembled by blending local contributions via the partition of unity :

$$\tilde{l}(x) = \sum_{i=1}^k \rho_i(\pi(x)) (c\tau \cdot \alpha_i \Phi_i f_i(x) + \pi(x)) \quad (12)$$

where $c \in (0, 1)$ is a scaling parameter ensuring non-intersection of fibers.

This construction smoothly interpolates the aligned local fiber embeddings around the base, yielding a continuous global representation that preserves both topology and local geometry.

5 Didactic Illustration: The Cylinder in $\mathbb{R}^3$

To illustrate the mechanism of the *FibeRed* algorithm with full visual fidelity, we consider a

system of 3 sensors (or pixels) as in the section 2 but with random white noise at each observation, defining an ambient space $\mathbb{R}^3$. Unlike high-dimensional examples where visualization requires projection, here Figure 6 represents the exact position of the data points in the space of pixel intensities (I_1, I_2, I_3).

The data forms a twisted cylinder structure composed of two elements: a global cyclic Base (S^1) representing a periodic phenomenon, and a local Fiber ($\mathbb{R}$) representing local variations.

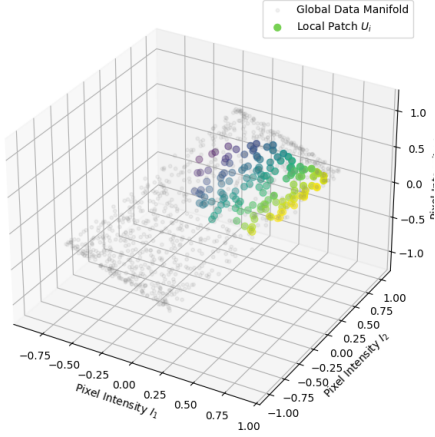


Figure 6: Ambient Space $\mathbb{R}^3$ (Raw Pixel Values)

1. Global Topology (Figure 6). Using persistent cohomology, we identify the circular topology and assign a base coordinate θ to each point. While we know where we are on the circle, we do not yet know the orientation of the cylinder’s thickness (the fiber) within the 3D space.
2. Local Geometry (Figure 7a). The algorithm selects a local patch of data (colored points in Fig. 6) and flattens it into a 2D Local Tangent Space. Inside this local chart, we identify two key directions. First, the Base Direction (Red Arrow), which corresponds to the direction of maximum variation of the topological coordinate θ . Second, the Fiber Direction (Blue Arrow), which is the subspace strictly orthogonal (the Kernel) to the Base.
3. Disentanglement (Figure 7b). By projecting the local data onto the Fiber Direction (the blue arrow), we extract the fiber coordinate f . The final plot maps the known θ against the extracted f , successfully unrolling the cylinder

into a flat rectangular domain without crushing its geometry.

It is important to understand that this illustrative example represents a trivial vector bundle, meaning that the global topology does not exhibit twists (unlike a Möbius strip). Consequently, all fibers are naturally aligned globally, and the transition matrices Ω_{ij} described in the general FibeRed algorithm essentially reduce to identity matrices. While this hugely simplifies the global alignment problem by removing the need for complex synchronization, the core conceptual approach, extracting the fiber locally as the kernel of the base projection, remains exactly the same.

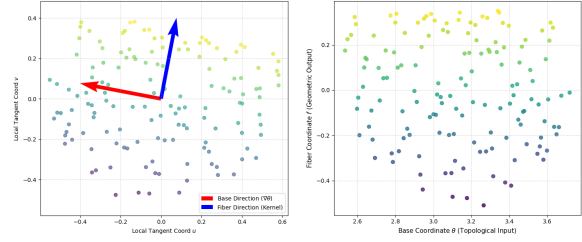


Figure 7: Visualization of the Tangent space (left image) and of the Disentangled representation E (right image).

6 Limitations.

As briefly noted at the end of the paper, this algorithm has several limitations, most of which arise from the assumption that the complex data lie in the total space of a vector bundle.

A major limitation of the algorithm is that its success depends on the first step, finding a good cover of the data. This step can fail if even a small cluster of points does not lie on the sampled manifold.

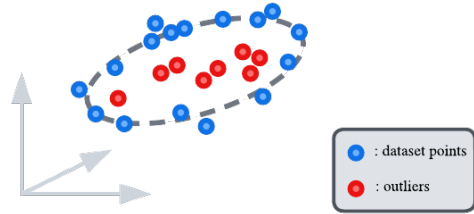


Figure 8: Schematic of an example dataset. The blue points represents the dataset points, a circularity is clearly visible. The red points are outliers and make the circularity less obvious.

To evaluate how the algorithm performs under such conditions, we introduce noise into the data

and add outliers outside the manifold. In the following section, we use the cylinder dataset provided by the authors, apply Gaussian noise, and then introduce said outliers.

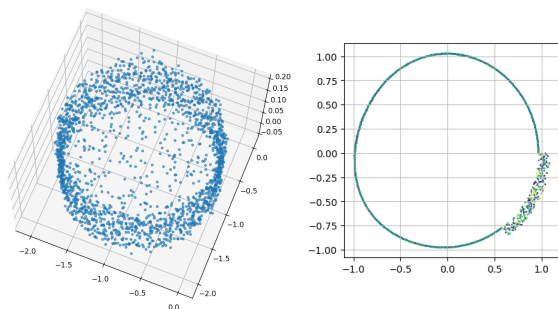


Figure 9: Visualization of dataset and the results of the FibeRed algorithm. In the left image, we can see the dataset where the circularity of the cylinder is clearly visible but altered. In the right image the FibeRed algorithm struggles to plot correctly the topology.

It is possible to compare FibeRed with a very well known algorithm, t -SNE.

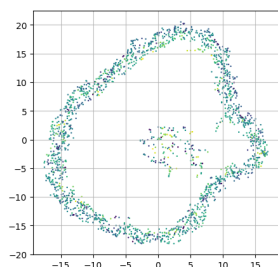


Figure 10: Visualization of the results of t -SNE with perplexity 30 on the same dataset.

While a superficial visual assessment might suggest that FibeRed outperforms t -SNE by recovering a clean circular structure, a deeper analysis reveals a limitation regarding the handling of outliers.

The FibeRed algorithm operates under a strong topological prior: it assumes that every point belongs to a bundle structure and computes a base coordinate $\theta = \pi(x)$ for the entire dataset. However, for outliers located in the hollow center of the cylinder, far from the manifold, this angular coordinate is ill-defined. By design, FibeRed projects these points onto the learned fiber structure, effectively "crushing" the outliers onto the circumference of the circle. This forced projection creates the density discontinuities observed

in the embedding (Figure 9b), where noise becomes indistinguishable from the signal.

In contrast, t -SNE, which prioritizes the preservation of local metric neighborhoods, successfully preserves the spatial separation between the manifold and the added noise in the center. As shown in Figure 10, it clusters the outliers in the center, reflecting their true topological position relative to the cylinder. Consequently, while FibeRed excels at reconstruction under strict manifold assumptions, t -SNE provides a more faithful representation of the actual data distribution, preserving the existence of outliers rather than assimilating them into the topology.

7 Conclusion.

In this project, we presented a detailed study of the paper **FibeRed: Fiberwise Dimensionality Reduction of Topologically Complex Data with Vector Bundles** by Luis Scoccola and Jose A. Perea, which introduces the FibeRed algorithm. This method relies on rather strong assumptions, such as requiring prior knowledge of a base map and assuming that the underlying manifold has no singularities. We demonstrated that, as suggested by the authors, the presence of outliers can significantly affect the algorithm's performance and its embedding fidelity regarding the true data distribution. Through relatively simple examples, we aimed to provide the reader with an intuition of the mathematical tools used in this algorithm and to explain why they are efficient in specific contexts. Ultimately, this method allows certain datasets to be represented in very low dimensions, drastically reducing complexity.

References

1. Luis Scoccola and Jose A. Perea. FibeRed: Fiberwise Dimensionality Reduction of Topologically Complex Data with Vector Bundles. In: *Proceedings of the 39th International Symposium on Computational Geometry (SoCG 2023)*, LIPIcs, vol. 258, pp. 56:1–56:18, 2023. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.
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